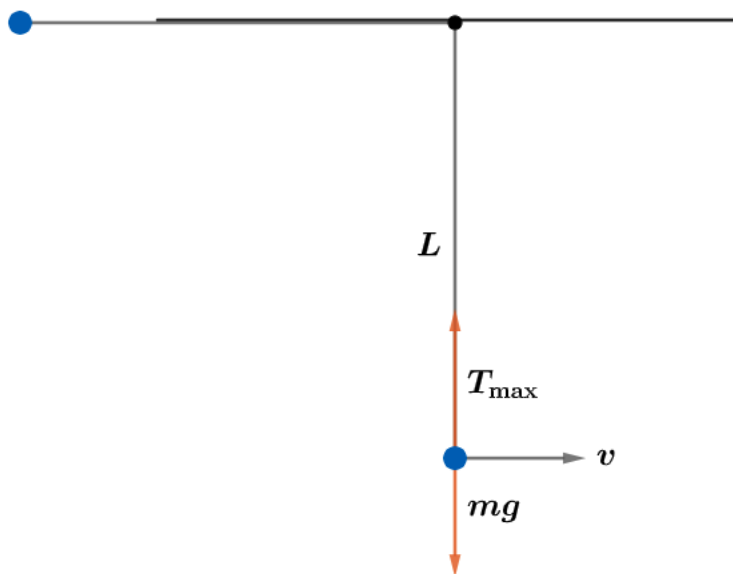


2022A F=ma Exam: Problem 6

Kevin S. Huang



The tension is maximized at the bottom of the bob's swing. By conservation of energy,

$$mgL = \frac{1}{2}mv^2$$
$$v^2 = 2gL$$

The tension and weight provide the centripetal acceleration, so

$$T_{\max} - mg = \frac{mv^2}{L} = 2mg$$
$$T_{\max} = 3mg$$

and the answer is D.